

Instalación de Docker

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1.Instalación de Docker

Instalación en Ubuntu:

sudo apt install docker.io

```
root@pablosg2526-VirtualBox:/home/pablosg2526# sudo apt install docker.io
```

Descarga de una imagen:

sudo docker pull hello-world

```
root@pablosg2526-VirtualBox:/home/pablosg2526# sudo docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
Digest: sha256:05813aedc15fb7b4d732e1be879d3252c1c9c25d885824f6295cab4538cb85cd
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
```

Ejecución de una imagen:

sudo docker run hello-world

```
root@pablosg2526-VirtualBox:/home/pablosg2526# sudo docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

Uso de docker sin ser root. se añade el usuario de Ubuntu dentro del grupo docker

sudo usermod -aG docker pablosg2526 (hay que cerrar la sesión y volver a entrar para que funcione; si no va, reiniciar Ubuntu)

```
root@pablosg2526-VirtualBox:/home/pablosg2526# sudo usermod -aG docker pablosg2526
```

Versión de docker instalada

sudo docker version

```

root@pablosg2526-VirtualBox:/home/pablosg2526# sudo docker version
Client:
 Version:           28.2.2
 API version:       1.50
 Go version:        go1.23.1
 Git commit:        28.2.2-0ubuntu1~24.04.1
 Built:            Wed Sep 10 14:16:39 2025
 OS/Arch:          linux/amd64
 Context:          default

Server:
 Engine:
  Version:          28.2.2
  API version:      1.50 (minimum version 1.24)
  Go version:       go1.23.1
  Git commit:       28.2.2-0ubuntu1~24.04.1
  Built:            Wed Sep 10 14:16:39 2025
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          1.7.28
  GitCommit:
 runc:
  Version:          1.3.3-0ubuntu1~24.04.3
  GitCommit:
 docker-init:
  Version:          0.19.0
  GitCommit:

```

Información sobre docker

sudo docker info

```

root@pablosg2526-VirtualBox:/home/pablosg2526# sudo docker info
Client:
 Version:           28.2.2
 Context:          default
 Debug Mode: false

Server:
 Containers: 1
  Running: 0
  Paused: 0
  Stopped: 1
 Images: 1
 Server Version: 28.2.2
 Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: false
  Native Overlay Diff: true
  userxattr: false
 Logging Driver: json-file
 Cgroup Driver: systemd
 Cgroup Version: 2
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
 CDI spec directories:
  /etc/cdi
  /var/run/cdi
 Swarm: inactive
 Runtimes: io.containerd.runc.v2 runc

```

```

Default Runtime: runc
Init Binary: docker-init
containerd version:
runc version:
init version:
Security Options:
  apparmor
  seccomp
   Profile: builtin
  cgroupns
Kernel Version: 6.14.0-37-generic
Operating System: Ubuntu 24.04.3 LTS
OSType: linux
Architecture: x86_64
CPUs: 2
Total Memory: 3.825GiB
Name: pablog2526-VirtualBox
ID: 5eec5c28-cab0-4c7e-8adf-62e9a057a18a
Docker Root Dir: /var/lib/docker
Debug Mode: false
Experimental: false
Insecure Registries:
  ::1/128
  127.0.0.0/8
Live Restore Enabled: false

```

Búsqueda de una imagen en hub.docker.com

docker search httpd

```

root@pablog2526-VirtualBox:~/home/pablog2526# docker search httpd
NAME                DESCRIPTION                STARS     OFFICIAL
httpd                The Apache HTTP Server Project 4919      [OK]
manageiq/httpd      Container with httpd, built on CentOS for Ma... 1
paketobuildpacks/httpd 0
vulhub/httpd        0
openeuler/httpd     0
dockerpinata/httpd  1
centos/httpd        36
e2eteam/httpd       0
manasip/httpd       0
amd64/httpd         The Apache HTTP Server Project 1
futdryt/httpd       0
arm64v8/httpd       The Apache HTTP Server Project 11
signiant/httpd      httpd (apache2) base container with a custom... 0
ppc64le/httpd       The Apache HTTP Server Project 0
9af925e7043/httpd   0
arm32v7/httpd       The Apache HTTP Server Project 11
tugboatqa/httpd     The Apache HTTP Server Project 0
s390x/httpd         The Apache HTTP Server Project 1
i386/httpd          The Apache HTTP Server Project 1
aerzas/httpd        HTTPD docker container image that requires n... 0
inventis/httpd       apache container with support for https only 0
armhf/httpd         The Apache HTTP Server Project 8
publici/httpd       httpd:latest 1
buzzerdev/httpd     Based on the official httpd image 0
vzwingmadomatic/httpd Service frontal de l'application de domotique 0

```

Búsqueda con filtro:

docker search ubuntu:24.04

```
root@pablosg2526-VirtualBox:/home/pablosg2526# docker search ubuntu:24.04
```

NAME	DESCRIPTION	STARS
ubuntu	Ubuntu is a Debian-based Linux operating sys...	17773
ubuntu/squid	Squid is a caching proxy for the Web. Long-t...	122
ubuntu/nginx	Nginx, a high-performance reverse proxy & we...	136
ubuntu/cortex	Cortex provides storage for Prometheus. Long...	4
ubuntu/kafka	Apache Kafka, a distributed event streaming ...	59
ubuntu/bind9	BIND 9 is a very flexible, full-featured DNS...	117
ubuntu/apache2	Apache, a secure & extensible open-source HT...	101
ubuntu/prometheus	Prometheus is a systems and service monitori...	78
ubuntu/zookeeper	ZooKeeper maintains configuration informatio...	14
ubuntu/jre	Chiseled Java runtime based on Ubuntu. Long...	23
ubuntu/mysql	MySQL open source fast, stable, multi-thread...	72
ubuntu/postgres	PostgreSQL is an open source object-relatio...	42
ubuntu/dotnet-aspnet	Chiselled Ubuntu runtime image for ASP.NET a...	26
ubuntu/python	A chiselled Ubuntu rock with the Python runt...	30
ubuntu/redis	Redis, an open source key-value store. Long...	24
ubuntu/dotnet-deps	Chiselled Ubuntu for self-contained .NET & A...	16
ubuntu/dotnet-runtime	Chiselled Ubuntu runtime image for .NET apps...	24
ubuntu/grafana	Grafana, a feature rich metrics dashboard & ...	12
ubuntu/memcached	Memcached, in-memory keyvalue store for smal...	6
ubuntu/prometheus-alertmanager	Alertmanager handles client alerts from Prom...	10
ubuntu/cassandra	Cassandra, an open source NoSQL distributed ...	3
ubuntu/mlflow	MLFlow: for managing the machine learning li...	6
ubuntu/loki	Grafana Loki, a log aggregation system like ...	2
ubuntu/telegraf	Telegraf collects, processes, aggregates & w...	4
ubuntu/chiselled-jre	[MOVED TO ubuntu/jre] Chiselled JRE: distrol...	3

Descargamos httpd y ubuntu

docker pull httpd

```
root@pablosg2526-VirtualBox:/home/pablosg2526# docker pull httpd
Using default tag: latest
latest: Pulling from library/httpd
119d43eec815: Pull complete
e0f36fcb159c: Pull complete
4f4fb700ef54: Pull complete
825b27552744: Pull complete
5e136ef74846: Pull complete
a062e9825802: Pull complete
Digest: sha256:dd178595edd6d4f49296f62f9587238db2cd1045adfff6fccc15a6c4d08f5d
Status: Downloaded newer image for httpd:latest
docker.io/library/httpd:latest
```

docker pull ubuntu

```
root@pablosg2526-VirtualBox:/home/pablosg2526# docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
a3629ac5b9f4: Pull complete
Digest: sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```

Ver las imágenes descargadas

docker images

Ver los contenedores en ejecución

docker ps -a

Si se ejecuta una imagen sin más, el contenedor puede arrancar y terminar sin más:

`docker run Ubuntu`

Para hacer un comando con ubuntu

`docker run ubuntu ls`

Lo hace y termina.

Para interactuar con un contenedor, se usan las opciones `-it`

`docker run -it ubuntu bash`

Aparecerá el identificador del contenedor:

Con otra terminal podemos hacer

`docker ps -a` y ver los estados de los contenedores que se han ejecutado hasta el momento:

Si se ejecuta un comando dentro del contenedor, como `mkdir pruebas`, lo hace, pero cuando termine el contenedor no recordará nada (no es persistente)

Salida de un contenedor:

`exit`

Borrado de un contenedor de la lista que sale con `docker ps -a`

`docker rm id_contenedor` Ej: `docker rm a5`